

Python Programming - Assignment-1

24BCA095

1. Store name, city and course in variables and display them

```
name = "Jeet"  
city = "Anand"  
course = "BCA"  
print(name)  
print(city)  
print(course)
```

```
Jeet  
Anand  
BCA
```

2. Create int, float, string and bool variables and display values and datatypes

```
a = 10  
b = 3.14  
c = "Hello"  
d = True  
print(a, type(a))  
print(b, type(b))  
print(c, type(c))  
print(d, type(d))  
  
10 <class 'int'>  
3.14 <class 'float'>  
Hello <class 'str'>  
True <class 'bool'>
```

3. Demonstrate dynamic typing

```
x = 100  
print(x, type(x))  
x = "Python"  
print(x, type(x))  
  
100 <class 'int'>  
Python <class 'str'>
```

4. Accept name and birth year, calculate age and print

```
import datetime  
name = input("Enter your name: ")
```

```
birth_year = int(input("Enter your birth year: "))
current_year = datetime.datetime.now().year
age = current_year - birth_year
print(name, age)
```

Jeet 20

5. Assign the same value to multiple variables and display them

```
a = b = c = 50
print(a)
print(b)
print(c)
```

50
50
50

6. Accept length and breadth of a rectangle and calculate area

```
length = float(input("Enter length: "))
breadth = float(input("Enter breadth: "))
area = length * breadth
print("Area of rectangle:", area)
```

Area of rectangle: 120.0

7. Accept radius of a circle and calculate area

```
radius = float(input("Enter radius: "))
area = 3.14 * radius * radius
print("Area of circle:", area)
```

Area of circle: 314.0

8. Perform all arithmetic operations on two numbers

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
print("Addition:", a + b)
print("Subtraction:", a - b)
print("Multiplication:", a * b)
print("Division:", a / b)
print("Modulus:", a % b)
print("Floor Division:", a // b)
print("Exponent:", a ** b)
```

Addition: 5
Subtraction: -1
Multiplication: 6

```
Division: 0.6666666666666666
Modulus: 2
Floor Division: 0
Exponent: 8
```

9. Calculate Simple Interest

```
p = float(input("Enter principal: "))
r = float(input("Enter rate: "))
t = float(input("Enter time: "))
si = (p * r * t) / 100
print("Simple Interest:", si)

Simple Interest: 15178.2
```

10. Accept Name, Course and Semester and display in formatted manner

```
name = input("Enter name: ")
course = input("Enter course: ")
semester = input("Enter semester: ")
print("Name:", name)
print("Course:", course)
print("Semester:", semester)

Name: JP
Course: BCA
Semester: 5
```

11. Check exam eligibility (Attendance >= 75 AND Internal Marks >= 20)

```
attendance = int(input("Enter attendance: "))
internal = int(input("Enter internal marks: "))
result = attendance >= 75 and internal >= 20
print(result)

False
```

12. Swap two numbers without using a third variable

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
a, b = b, a
print("First number:", a)
print("Second number:", b)

First number: 10
Second number: 1
```

13. Check whether a character exists in a string using Membership Operator

```
s = input("Enter a string: ")
ch = input("Enter a character: ")
print(ch in s)
```

False

14. Calculate square and cube using exponentiation operator

```
n = int(input("Enter a number: "))
print("Square:", n ** 2)
print("Cube:", n ** 3)
```

Square: 100
Cube: 1000

15. Classify a number as Positive, Negative or Zero

```
n = int(input("Enter a number: "))
if n > 0:
    print("Positive")
elif n < 0:
    print("Negative")
else:
    print("Zero")
```

Positive

16. If number is divisible by 5, print 'Divisible by 5'

```
n = int(input("Enter a number: "))
if n % 5 == 0:
    print("Divisible by 5")
```

Divisible by 5

17. Accept three numbers and print the largest

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
c = int(input("Enter third number: "))
if a >= b and a >= c:
    print("Largest:", a)
elif b >= a and b >= c:
    print("Largest:", b)
else:
    print("Largest:", c)
```

Largest: 20

18. Loan eligibility using nested if

```
name = input("Enter name: ")
phone = input("Enter phone no: ")
age = int(input("Enter age: "))
salary = float(input("Enter monthly salary: "))
if age >= 18:
    if salary >= 25000:
        print("Loan Approved")
    else:
        print("Loan Rejected: Low Salary")
else:
    print("Not Eligible Age Below 18")
```

Loan Approved

19. Calculate Profit or Loss

```
cp = float(input("Enter cost price: "))
sp = float(input("Enter selling price: "))
if sp > cp:
    print("Profit:", sp - cp)
elif sp < cp:
    print("Loss:", cp - sp)
else:
    print("No Profit No Loss")
```

Profit: 700.0

20. Print all numbers from 1 to N using while loop

```
n = int(input("Enter a number: "))
i = 1
while i <= n:
    print(i)
    i += 1
```

1
2
3
4
5
6
7
8
9
10

21. Sum of all numbers from 1 to N using while loop

```
n = int(input("Enter a number: "))
i = 1
total = 0
while i <= n:
    total += i
    i += 1
print("Sum:", total)
```

Sum: 55

22. Print multiplication table from 1 to 10 using for loop

```
n = int(input("Enter a number: "))
for i in range(1, 11):
    print(n, "x", i, "=", n * i)
```

```
10 x 1 = 10
10 x 2 = 20
10 x 3 = 30
10 x 4 = 40
10 x 5 = 50
10 x 6 = 60
10 x 7 = 70
10 x 8 = 80
10 x 9 = 90
10 x 10 = 100
```

23. Sum of all even numbers from 1 to N using for loop

```
n = int(input("Enter a number: "))
total = 0
for i in range(1, n + 1):
    if i % 2 == 0:
        total += i
print("Sum of even numbers:", total)
```

Sum of even numbers: 30

24. Calculate factorial of N using for loop

```
n = int(input("Enter a number: "))
fact = 1
for i in range(1, n + 1):
    fact *= i
print("Factorial:", fact)
```

Factorial: 3628800

25. Print Fibonacci series using while loop

```
n = int(input("Enter number of terms: "))
a = 0
b = 1
count = 0
while count < n:
    print(a, end=" ")
    c = a + b
    a = b
    b = c
    count += 1
```

0 1 1

26. Check if a number is Prime or Not Prime using for loop

```
n = int(input("Enter a number: "))
if n > 1:
    prime = True
    for i in range(2, n):
        if n % i == 0:
            prime = False
            break
    if prime:
        print("Prime")
    else:
        print("Not Prime")
else:
    print("Not Prime")
```

Not Prime

27. Check whether a year is a Leap Year or Non-Leap Year

```
year = int(input("Enter a year: "))
if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
    print("Leap Year")
else:
    print("Non-Leap Year")
```

Leap Year

28. Print all the factors of a number

```
n = int(input("Enter number: "))
print("Factors of", n, "are:", end=" ")
for i in range(1, n + 1):
    if n % i == 0:
        print(i, end=" ")
```

Factors of 10 are: 1 2 5 10